

**Project Title:** Shepherd (Smart Holistic Engine for Production, Handling, Editing, Routing, and Delivery)

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## 1. Project Overview

Shepherd is a local-first, AI-assisted production system designed for churches and live event media teams. It acts as a bridge between live production tools (Reaper, OBS, Yamaha console) and post-production workflows (editing, export, publishing). Shepherd automates transcription, segment detection, content generation, and media organization.

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## 2. Core Goals

- Integrate live audio/video capture (via Dante, Reaper, OBS).
  - Automatically organize, tag, and label projects by event/date/type.
  - Generate multi-format content outputs (sermons, podcasts, reels, highlight clips).
  - Leverage local AI models for transcription, summarization, and tagging.
  - Minimize human error and streamline the church media workflow.
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## 3. System Architecture

### Frontend (Desktop App)

- **Framework:** Electron + Next.js (React UI within desktop shell)
- **UI Elements:**
  - Dashboard (Project overview, current sessions, quick actions)
  - Project Manager (Folder view, metadata, export paths)
  - Transcription Panel (Whisper outputs, timestamps, speaker detection)
  - Segment Editor (Visual timeline, auto-detected sections like Sermon/Worship/Announcements)
  - AI Assistant (Chat or side panel powered by local LLM)
  - Export Manager (Choose export type: Sermon, Reels, Podcast, etc.)

### Backend (Local Server)

- **Framework:** FastAPI (Python)
- **Responsibilities:**
  - Handle Reaper/OBS/FFmpeg automation
  - Process audio/video files
  - Run local Whisper transcription
  - Connect with local LLM (Mistral/Llama) for summaries, titles, and descriptions
  - Maintain SQLite/Postgres database for projects and metadata
  - Expose REST endpoints for Electron frontend

### AI & Processing Layer

- **Speech-to-Text:** Whisper (local)

- **Language Understanding:** Llama 3.1 8B or Mistral 7B (GGUF quantized)
  - **Search/Indexing:** Chroma or Qdrant (vector database)
  - **Automation:** FFmpeg + ReaScript (Reaper scripting) + OBS WebSockets
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## 4. Minimum Usable Version (MVP)

The MVP focuses on automating repetitive tasks for audio and media management.

**MVP Features:** 1. Project creation & folder structure automation. 2. Local transcription using Whisper (auto audio import from Reaper session folder). 3. Sermon segmentation based on silence/energy detection. 4. Summary, title, and description generation using local LLM. 5. Export pipeline: - Audio cleanup (normalize, noise reduction) - Export sermon mix + podcast version 6. Dashboard with basic project management (list, rename, archive).

**MVP Stack:** - **Frontend:** Electron + Next.js - **Backend:** FastAPI + Python scripts - **Storage:** SQLite + local file system - **Models:** Whisper (medium), Llama/Mistral (7B GGUF) - **Automation:** FFmpeg, ReaScript (for Reaper control)

**Hardware Fit:** GTX 1080 (8GB VRAM) running quantized models locally.

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## 5. Future Version (v2.0 and beyond)

### v2.0 Goals:

- Multi-user collaboration with local network sync (e.g., media team access)
- Scene detection in video (AI-based visual segmentation)
- Auto-thumbnail generation (via vision model)
- AI audio cleanup (dereverb, noise, spectral gating)
- Sermon database with searchable archives (vector search)
- Analytics dashboard (viewer engagement, sermon trends)
- Smart scheduling & reminders for editing/export deadlines

### v3.0 Goals:

- Real-time stream companion: AI-assisted audio monitoring and scene switching suggestions for OBS.
  - Integration with cloud storage and YouTube API for direct upload.
  - Fine-tuned LLM on church-specific dataset for improved contextual summaries.
  - Web/mobile dashboard for remote management.
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## 6. Data Flow (Simplified)

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[Yamaha Mixer] → [Dante → Reaper Multitrack] → [Whisper STT] → [AI Segmentation + Labeling]
→ [LLM Summarization + Metadata] → [Export via FFmpeg] → [Dashboard Display + Archive]
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## 7. Development Phases

**Phase 1: Infrastructure Setup** - Folder structure automation (already built via PowerShell) - Install REAPER, OBS, FFmpeg, Whisper, local LLM backend - FastAPI + Electron scaffolding

**Phase 2: Core AI Integration** - Whisper integration with FastAPI endpoints - LLM-based text generation - Audio segmentation pipeline

**Phase 3: User Interface & Automation** - Build dashboard UI in Electron/Next.js - Integrate project creation, Whisper transcription, and export flow

**Phase 4: Smart Automation** - Auto export + file routing - OBS & Reaper scripting automation

**Phase 5: Advanced AI Layer** - Scene detection, thumbnail generation, metadata analytics

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## 8. Example Workflow

1. Create new project → folder + metadata auto-generated.
2. Import or record audio via Reaper → save to project folder.
3. Whisper transcribes audio → auto-detects segments.
4. LLM summarizes + titles each segment.
5. Export pipeline cleans audio and creates deliverables.
6. Dashboard displays results; user can review, edit, and publish.

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## 9. Licensing & Cost Notes

- **Free/Local:** Whisper, Llama, Mistral (quantized), FFmpeg, FastAPI, Electron, Next.js.
- **Optional Paid:** DaVinci Resolve, premium plugins, GPU cloud scaling.
- **Estimated local cost:** \$0 per month (excluding electricity & hardware).

## 10. Vision Summary

Shepherd transforms a high-performance church media setup into a unified creative system. It automates technical drudgery, structures the creative process, and enables media teams to focus on storytelling and ministry impact rather than cables and exports.

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**Next Steps:** 1. Initialize FastAPI backend. 2. Scaffold Electron + Next.js UI. 3. Connect Whisper + LLM integration. 4. Prototype local dashboard for transcription and summaries.

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